

LLM-Driven Quadruped Robot Inspection System with Human-in-the-Loop Control via Model Context Protocol

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Abstract—Industrial inspection using quadruped robots presents a challenge in balancing operational autonomy with the adaptability needed for dynamic environments. Existing approaches are either fully teleoperated—imposing high cognitive load on operators—or fully autonomous with rigid, pre-programmed plans that cannot respond to semantic human commands or unexpected field conditions. This paper presents a distributed inspection system integrating a Large Multimodal Model (LMM), specifically Google Gemini, as an agentic orchestrator for a quadruped robot (DEEPRobotics Jueying Lite3 Pro). The architecture employs the Model Context Protocol (MCP) to bridge the LLM agent with a ROS-based navigation stack, enabling semantic instruction processing, autonomous navigation, and SOP-based visual inspection. A stepwise Human-in-the-Loop (HITL) mechanism—facilitated through webhook callbacks—ensures operator validation at each mission stage while preserving system adaptability. The system further applies Retrieval-Augmented Generation (RAG) to ground LLM decisions in structured operational data (waypoints and SOP documents). Experimental results demonstrate the system’s ability to autonomously plan and execute multi-waypoint inspection missions from natural language commands with accurate SOP compliance checking.

Index Terms—quadruped robot, large language model, agentic AI, human-in-the-loop, model context protocol, ROS, autonomous inspection

I. INTRODUCTION

The rapid growth of autonomous mobile robotics—projected to reach USD 9.56 billion by 2030 [1]—reflects increasing demand for automation in high-risk industrial environments. Quadruped robots have emerged as particularly capable platforms for such environments due to their superior terrain adaptability compared to wheeled alternatives [2]. However, despite this mobility advantage, the deployment of quadruped robots for industrial inspection remains operationally challenging.

Current inspection paradigms fall into two extremes. *Teleoperation* reduces operator risk but imposes continuous cognitive demand: operators must simultaneously interpret raw telemetry and visual feeds while issuing motor commands [3]. *Fully autonomous* approaches address this through deterministic navigation to pre-programmed waypoints, as demonstrated by

Darmawan [4], but remain inflexible—any change in inspection scope requires code-level reprogramming.

Initial attempts at integrating Large Language Models (LLMs) for robot control, such as that proposed by Mahendra [5], offer natural language flexibility but generate and execute complete action plans non-interactively. This “batch-execution” approach provides no mechanism for operator intervention mid-mission, which is critical in dynamic industrial settings where conditions may change unexpectedly [6].

This paper addresses these limitations through three contributions:

- 1) A distributed agentic architecture connecting a multimodal LLM to a ROS-based robot controller via the Model Context Protocol (MCP), enabling semantic command interpretation and tool-based execution.
- 2) A stepwise Human-in-the-Loop (HITL) mechanism using webhook callbacks that pauses execution after each robot action, awaiting operator validation before proceeding.
- 3) A Retrieval-Augmented Generation (RAG) pipeline that grounds LLM decisions in structured inspection data (waypoint coordinates and SOP documents) retrieved on demand.

II. RELATED WORK

A. LLM Integration in Robotics

The application of LLMs as high-level robot planners has been extensively explored. SayCan [8] grounds language commands in robot affordances, while PaLM-E [9] demonstrates multimodal understanding for robotic manipulation. These works primarily focus on manipulation tasks; our work extends the paradigm to long-horizon autonomous inspection in real factory environments.

B. Human-in-the-Loop Robot Control

HITL mechanisms are well-established in semi-autonomous robotics [7]. However, most implementations focus on teleoperation adjustment rather than LLM-orchestrated step-validation. Our approach treats operator confirmation as a

first-class event in the agentic loop, architecturally enforced through asynchronous webhook callbacks.

C. RAG for Operational Grounding

Retrieval-Augmented Generation [10] has been successfully applied to ground LLM outputs in domain-specific corpora. We adapt this to robotics by retrieving structured spatial data (waypoints) and procedural documents (SOPs) at tool-execution time, rather than at prompt construction time.

III. SYSTEM ARCHITECTURE

A. Overview

The system adopts a distributed architecture comprising four loosely coupled components, as illustrated in Fig. 1. Communication between components follows explicit protocols to ensure modularity and replaceability.

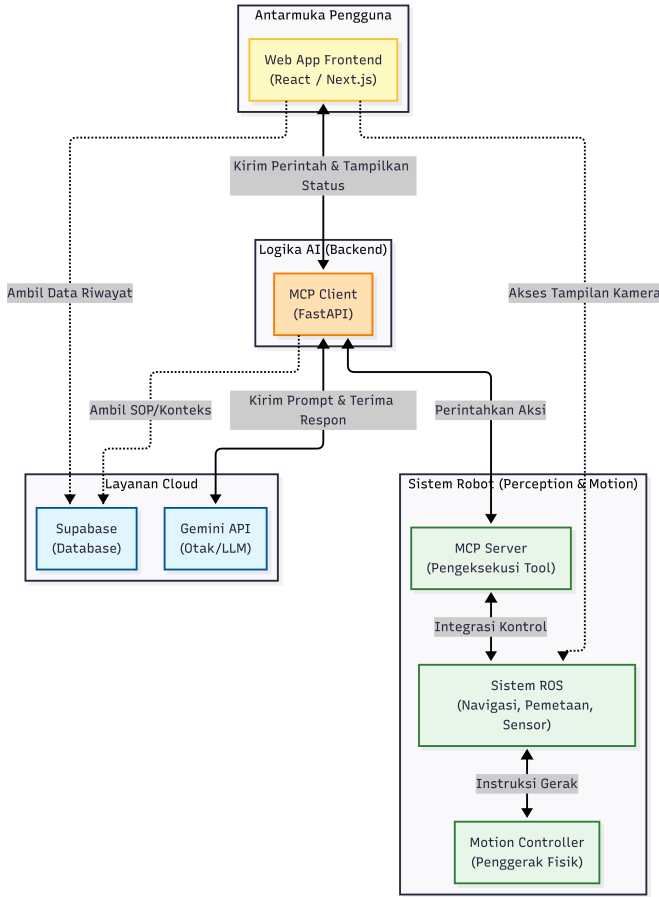


Fig. 1. System architecture. The LLM agent (MCP Client) bridges the operator interface and the robot-side MCP Server via the Model Context Protocol. Cloud services (Gemini API and Supabase) are accessed through the internet.

- **Web Dashboard (robodog-web):** A Next.js application serving as the operator control station. It communicates with the backend via REST API for chat and data management. Real-time telemetry (robot pose, occupancy map) is streamed directly from the ROS ecosystem via

rosbridge WebSocket. Camera video is delivered over WebRTC.

- **LLM Agent / MCP Client (robodog-client):** A FastAPI service that acts as the agentic middleware. It holds the system prompt defining the inspection workflow, manages conversation history in Supabase, and implements the agentic loop: forward prompts to Gemini, intercept *function_call* responses, dispatch tool calls to the MCP Server, and inject results back into the LLM context.
- **MCP Server (robodog-server):** A FastMCP HTTP server running on the robot's onboard AI computer (NVIDIA Jetson Xavier NX). It exposes six tools (Table II) and handles all robot-side execution including ROS interaction and database retrieval.
- **Cloud Services:** Google Gemini API provides LLM inference. Supabase provides PostgreSQL database storage (sessions, messages, locations, waypoints, objects) and file storage (SOP documents, captured images, maps).

B. Communication Protocols

Table I summarizes the inter-component communication.

TABLE I
COMMUNICATION PROTOCOLS BETWEEN SYSTEM COMPONENTS

Link	Protocol	Purpose
Web ↔ Backend	REST / HTTP	Chat messages, CRUD
Web ↔ ROS	WebSocket (rosbridge)	Telemetry, maps
Web ↔ Robot camera	WebRTC	Live video
Backend ↔ MCP Server	MCP over HTTP/SSE	Tool invocation
MCP Server ↔ ROS	rospy (internal)	cmd_vel, move_base
Robot → Backend	HTTP POST (webhook)	Action completion report
Backend ↔ Gemini	Google GenAI SDK	LLM inference
All ↔ Supabase	REST API / SDK	Database & storage

C. Hardware Platform

The robot platform is the DEEPRobotics Jueying Lite3 Pro [14], a 12-DOF quadruped (mass 12.7 kg, 610×370×445 mm standing), augmented with a Leishen C16 16-channel 3D LiDAR [15]. The onboard AI computer is an NVIDIA Jetson Xavier NX, which runs all software components (ROS stack, MCP Server, MCP Client, and web application) as a single-host deployment managed by *systemd* services and Docker Compose. Remote access is provided through Tailscale VPN, eliminating the need for a shared local network.

IV. SYSTEM DESIGN

A. MCP Tool Definitions

The MCP Server exposes six tools that collectively span the full capability surface of the robot, as detailed in Table II. Tool schemas are converted from MCP format to Gemini *FunctionDeclaration* format at runtime.

B. Agentic Loop and HITL Mechanism

The inspection workflow is orchestrated by an agentic loop inside the MCP Client, as shown in Fig. 2. The loop operates as follows:

TABLE II
MCP TOOLS EXPOSED BY THE ROBOT SERVER

Tool	Description
move	Manual open-loop velocity control (speed, duration)
navigate_to_waypoint	Autonomous navigation via <i>move_base</i> Action (x, y, θ)
get_object_waypoints	Hierarchical DB query: location $\rightarrow$ object $\rightarrow$ waypoints
list_sop_files	List SOP files in Supabase Storage
get_sop_file	Download and return SOP document bytes
capture_and_upload_image	Capture camera frame via RTSP, upload to Supabase, return URL

- 1) The operator issues a natural language command (e.g., “Inspect all fire extinguishers in Zone A”).
- 2) The backend loads conversation history from Supabase and forwards the full context to Gemini API.
- 3) Gemini responds with a *function_call*. The backend dispatches the call to the MCP Server.
- 4) If the tool involves data retrieval (e.g., *get_object_waypoints*, *get_sop_file*), the MCP Server queries Supabase. File bytes (SOP PDFs, images) are injected directly into the Gemini context as `Part.from_bytes` for multimodal analysis.
- 5) If the tool involves robot action (e.g., *navigate_to_waypoint*), execution is asynchronous: the MCP Server initiates the *move_base* goal and immediately returns. Upon goal completion (or failure), the ROS controller fires an HTTP POST webhook to the MCP Client.
- 6) The webhook triggers a new Gemini invocation with the action result. Gemini presents findings to the operator and **pauses** for explicit confirmation before proceeding to the next waypoint (HITL gate).
- 7) The loop repeats until the full inspection plan is executed or the operator aborts.

C. RAG-Based Operational Grounding

Rather than encoding inspection knowledge in the system prompt, the architecture uses tool-time retrieval. When Gemini identifies an inspection target, it calls *get_object_waypoints* to fetch spatial coordinates from the database and *get_sop_file* to retrieve the relevant SOP document. This approach ensures that:

- The LLM always operates on the latest operational data without redeployment.
- Inspection procedures can be updated by operators (uploading new SOP files) without modifying any code.
- Hallucination risk is reduced as spatial data is retrieved from a ground-truth database rather than generated.

D. Navigation Stack

The ROS Navigation Stack is configured with a Dijkstra-based global planner and a Timed Elastic Band (TEB) local

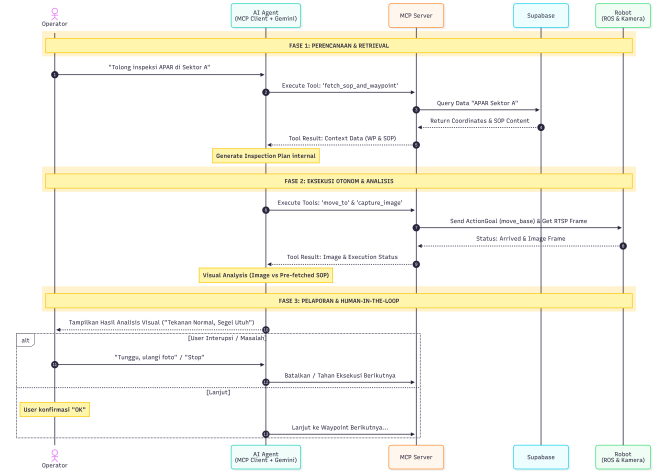


Fig. 2. Sequence diagram of the inspection workflow showing the three-phase agentic loop: (1) Retrieval & Planning, (2) Autonomous Execution & Analysis, (3) HITL Reporting.

planner [11]. Localization is performed by HDL Localization using NDT scan matching [12] and Fast-GICP [13] against a 3D point cloud map built by Faster-LIO (LiDAR-Inertial Odometry). The 3D map is converted to a 2D occupancy grid via `pcd_2_gridmap` for use by the Navigation Stack.

V. IMPLEMENTATION

A. Deployment

All software components are deployed on the robot’s Jetson Xavier NX. Four *systemd* services manage the core stack: LiDAR driver, navigation (*move_base* + localization), MCP Server (port 8001), and MCP Client/Backend (port 8082). The web dashboard runs in a Docker Compose container (port 3000). Tailscale VPN provides a stable overlay network; the Supabase instance runs on the developer’s laptop, mimicking a cloud deployment without public infrastructure requirements.

B. System Prompt Engineering

The system prompt instructs Gemini to follow a strict inspection protocol:

- 1) Retrieve the relevant SOP document before any navigation.
- 2) Query waypoints for the target object.
- 3) Present a step-by-step inspection plan and *wait for operator approval*.
- 4) Execute waypoints sequentially, one at a time.
- 5) After each capture, analyze the image against the SOP and report findings.
- 6) Ask the operator whether to continue to the next waypoint.

This prompt-level enforcement of the HITL protocol is complemented by the architectural webhook mechanism.

C. Web Dashboard

The operator dashboard (Fig. 3) is a single-page Next.js application with eight functional pages. The primary *Inspection Hub* page features a split-screen layout: the left panel shows a live map (rendered from ROS topics via roslibjs) with the robot's real-time pose, and the right panel provides the LLM chat interface. Supporting pages provide management interfaces for spatial locations, inspection objects, SOP documents, captured image galleries, and session history.

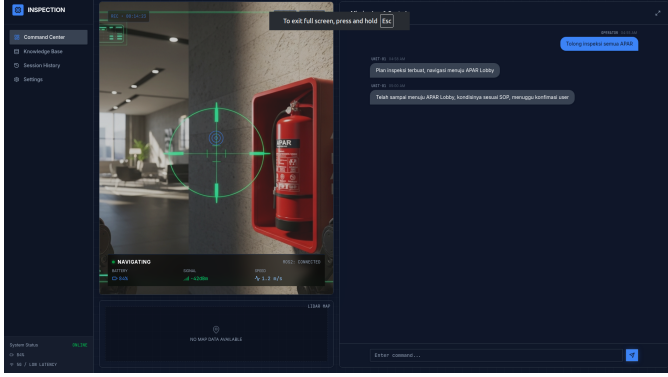


Fig. 3. Web dashboard Inspection Hub: split-screen layout with real-time navigation map (left) and LLM chat panel (right).

VI. EVALUATION

A. Experimental Setup

Experiments were conducted in an indoor laboratory environment replicating an industrial inspection scenario. The environment contains multiple inspection objects (fire extinguishers, electrical panels) distributed across a mapped area. The robot was tasked with inspecting all objects in a zone based on a single natural language command, without pre-programmed waypoint sequences.

B. Localization Accuracy

Localization performance was evaluated by comparing the robot's estimated pose from HDL Localization against ground-truth positions measured with a tape measure. [Results to be filled based on experimental data.]

C. Navigation Success Rate

Navigation success was measured as the fraction of waypoints reached within an acceptable tolerance radius. [Results to be filled based on experimental data.]

D. Inspection Workflow Accuracy

The system's ability to correctly identify inspection objects, retrieve appropriate SOPs, and generate accurate compliance assessments was evaluated by domain experts comparing the robot's reports against manual inspection records. [Results to be filled based on experimental data.]

E. Operator Workload

Operator cognitive load was assessed qualitatively through structured interviews comparing the proposed HITL system against teleoperation. Initial results indicate a significant reduction in continuous attention demand, as operators only need to evaluate presented findings rather than actively guide the robot.

VII. CONCLUSION

This paper presented an LLM-driven quadruped robot inspection system that addresses the limitations of both teleoperation and fully autonomous approaches. By combining the Model Context Protocol for structured tool invocation, webhook-based asynchronous reporting, and a stepwise Human-in-the-Loop validation mechanism, the system enables flexible and safe inspection missions from natural language commands. The RAG-based grounding approach ensures that LLM decisions are anchored to authoritative operational data, reducing hallucination risk in safety-critical inspection contexts. Future work will focus on extending the system to outdoor environments, integrating thermal imaging for anomaly detection, and conducting large-scale user studies to quantify operator workload reduction.

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